

1:

```
> db.students.find({
  dept: "CS",
  marks: { $gt: 80 }
})
< [
  {
    _id: 1,
    name: 'Ananya',
    age: 21,
    dept: 'CS',
    marks: 88,
    grades: [
      'A',
      'B',
      'A'
    ],
    address: {
      city: 'Chennai',
      pin: '600001'
    },
    active: true
  },
  {
    _id: 3,
    name: 'Meena',
    age: 20,
    dept: 'CS',
    marks: 95,
    grades: [
      'A',
      'A',
      'A'
    ],
    address: {
      city: 'Chennai',
      pin: '600002'
    },
    active: true
  }
]
```

2:

```
> db.students.find({
  marks: {
    $gte: 60,
    $lte: 85
  }
})
< [
  {
    _id: 2,
    name: 'Arjun',
    age: 23,
    dept: 'IT',
    marks: 74,
    grades: [
      'B',
      'C',
      'B'
    ],
    address: {
      city: 'Madurai',
      pin: '625001'
    },
    active: true
  },
  {
    _id: 4,
    name: 'Ravi',
    age: 22,
    dept: 'ECE',
    marks: 61,
    grades: [
      'C',
      'C',
      'B'
    ],
```

```
    address: {
      city: 'Trichy',
      pin: '620001'
    },
    active: false
  },
  {
    _id: 5,
    name: 'Kavya',
    age: 24,
    dept: 'IT',
    marks: 82,
    grades: [
      'A',
      'B',
      'B'
    ],
    address: {
      city: 'Coimbatore',
      pin: '641001'
    },
    active: true
  },
  {
    _id: 7,
    name: 'Priya',
    age: 22,
    dept: 'ECE',
    marks: 79,
    grades: [
      'B',
      'B',
      'A'
    ],
```

```
    address: {
      city: 'Coimbatore',
      pin: '641001'
    },
    active: true
  },
  {
    _id: 7,
    name: 'Priya',
    age: 22,
    dept: 'ECE',
    marks: 79,
    grades: [
      'B',
      'B',
      'A'
    ],
    address: {
      city: 'Chennai',
      pin: '600003'
    },
    active: true
  },
  {
    _id: 9,
    name: 'Rakha',
    age: 21,
    dept: 'CS',
    marks: 68,
    grades: [
      'B',
      'C',
      'B'
    ],
    address: {
      city: 'Trichy',
      pin: '620002'
    },
```

3:

```
> db.students.find({
  dept: { $in: ["IT", "ECE"] }
})
< [
  {
    _id: 2,
    name: 'Arjun',
    age: 23,
    dept: 'IT',
    marks: 74,
    grades: [
      'B',
      'C',
      'B'
    ],
    address: {
      city: 'Madurai',
      pin: '625001'
    },
    active: true
  },
  {
    _id: 4,
    name: 'Ravi',
    age: 22,
    dept: 'ECE',
    marks: 61,
    grades: [
      'C',
      'C',
      'B'
    ],
    address: {
      city: 'Trichy',
      pin: '620001'
    },
    active: true
  },
  {
    _id: 8,
    name: 'Suresh',
    age: 25,
    dept: 'IT',
    marks: 90,
    grades: [
      'A',
      'A',
      'B'
    ],
    address: {
      city: 'Madurai',
      pin: '625002'
    },
    active: false
  },
  {
    _id: 10,
    name: 'Vijay',
    age: 23,
    dept: 'ECE',
    marks: 45,
    grades: [
      'D',
      'F',
      'D'
    ],
    address: {
      city: 'Coimbatore',
      pin: '641002'
    },
    active: false
  }
]
```

4:

```
db.students.find(
  {
    dept: { $nin: ["CS"] }
  },
  {
    _id: 0,
    name: 1,
    dept: 1
  }
)
[
  {
    name: 'Arjun',
    dept: 'IT'
  }
  {
    name: 'Ravi',
    dept: 'ECE'
  }
  {
    name: 'Kavya',
    dept: 'IT'
  }
  {
    name: 'Priya',
    dept: 'ECE'
  }
  {
    name: 'Suresh',
    dept: 'IT'
  }
  {
    name: 'Vijay',
    dept: 'ECE'
  }
]
```

5:

```
> db.students.find(
  {
    dept: { $nin: ["CS"] }
  },
  {
    _id: 0,
    name: 1,
    dept: 1
  }
)
< {
  name: 'Arjun',
  dept: 'IT'
}
{
  name: 'Ravi',
  dept: 'ECE'
}
{
  name: 'Kavya',
  dept: 'IT'
}
{
  name: 'Priya',
  dept: 'ECE'
}
{
  name: 'Suresh',
  dept: 'IT'
}
{
  name: 'Vijay',
  dept: 'ECE'
}
```

```

> db.products.aggregate([
  {
    $group: {
      _id: "$category",
      totalRevenue: {
        $sum: {
          $multiply: ["$price", "$sold"]
        }
      }
    }
  }
])
< {
  _id: 'Books',
  totalRevenue: 629000
}
{
  _id: 'Electronics',
  totalRevenue: 16817000
}
{
  _id: 'Furniture',
  totalRevenue: 612000
}

```

6:

```

> db.students.aggregate([
  {
    $group: {
      _id: "$dept",
      totalStudents: { $sum: 1 },
      averageMarks: { $avg: "$marks" }
    }
  }
])
< {
  _id: 'ECE',
  totalStudents: 3,
  averageMarks: 61.666666666666664
}
{
  _id: 'CS',
  totalStudents: 4,
  averageMarks: 76.5
}
{
  _id: 'IT',
  totalStudents: 3,
  averageMarks: 82
}

```

7:

```
> db.students.updateMany(  
  {},  
  [  
    $unset: { scholarship: "" }  
  ]  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 10,  
  modifiedCount: 0,  
  upsertedCount: 0  
}
```

8:

```
> db.students.updateMany(  
  { dept: "CS" },  
  [  
    $set: { scholarship: false }  
  ]  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 4,  
  modifiedCount: 4,  
  upsertedCount: 0  
}
```

9:

```
> db.students.updateOne(  
  { _id: 6 },  
  [  
    $inc: { marks: 10 }  
  ]  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

10:

```

> db.students.find()
.sort({ marks: -1 })
.skip(2)
.limit(2)
< [
  {
    _id: 1,
    name: 'Ananya',
    age: 21,
    dept: 'CS',
    marks: 88,
    grades: [
      'A',
      'B',
      'A'
    ],
    address: {
      city: 'Chennai',
      pin: '600001'
    },
    active: true,
    scholarship: false
  },
  {
    _id: 5,
    name: 'Kavya',
    age: 24,
    dept: 'IT',
    marks: 82,
    grades: [
      'A',
      'B',
      'B'
    ],
    address: {
      city: 'Coimbatore',
      pin: '641001'
    },
    active: true
  }
]

```

11: }


```
db.employees.find({
  salary: { $gt: 50000 },
  experience: { $gt: 5 }
})
[
  {
    _id: ObjectId('6a26bce9015f1fd60bbfbd7e'),
    empId: 103,
    name: 'Rahul',
    age: 32,
    gender: 'Male',
    department: 'IT',
    salary: 75000,
    city: 'Mumbai',
    skills: [
      'Python',
      'MongoDB',
      'AWS'
    ],
    experience: 8,
    active: true
  }
  {
    _id: ObjectId('6a26bce9015f1fd60bbfbd80'),
    empId: 105,
    name: 'Karthik',
    age: 30,
    gender: 'Male',
    department: 'IT',
    salary: 68000,
    city: 'Chennai',
    skills: [
      'NodeJS',
      'MongoDB',
      'React'
    ],
    experience: 7,
    active: true
  }
]
```

12: }

```

> db.employees.find([
  $expr: {
    $gt: [
      { $size: "$skills" },
      2
    ]
  }
])
< [
  {
    _id: ObjectId('6a26bce9015f1fd60bbfbd7c'),
    empId: 101,
    name: 'Arun',
    age: 25,
    gender: 'Male',
    department: 'IT',
    salary: 45000,
    city: 'Chennai',
    skills: [
      'Java',
      'MongoDB',
      'Spring'
    ],
    experience: 3,
    active: true
  },
  {
    _id: ObjectId('6a26bce9015f1fd60bbfbd7e'),
    empId: 103,
    name: 'Rahul',
    age: 32,
    gender: 'Male',
    department: 'IT',
    salary: 75000,
    city: 'Mumbai',
    skills: [
      'Python',
      'MongoDB',
      'AWS'
    ],
    experience: 5,
    active: true
  }
]

```

13:

```

> db.employees.aggregate([
  {
    $group: {
      _id: "$department",
      avgSalary: { $avg: "$salary" }
    }
  },
  {
    $match: {
      avgSalary: { $gt: 50000 }
    }
  }
])
< [
  {
    _id: 'IT',
    avgSalary: 62666.666666666664
  },
  {
    _id: 'HR',
    avgSalary: 55000
  }
]

```

14:

```

> db.employees.aggregate([
  {
    $group: {
      _id: "$department",
      minSalary: { $min: "$salary" }
    }
  }
])
< [
  {
    _id: 'HR',
    minSalary: 55000
  },
  {
    _id: 'IT',
    minSalary: 45000
  },
  {
    _id: 'Finance',
    minSalary: 50000
  },
  {
    _id: 'Marketing',
    minSalary: 42000
  }
]

```

15:

```
> db.employees.updateMany(  
  {},  
  [  
    $rename: { city: "location" }  
  ]  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 6,  
  modifiedCount: 6,  
  upsertedCount: 0  
}
```

16:


```
> db.students.updateMany(  
  { active: false },  
  {  
    $set: { active: true }  
  }  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 3,  
  modifiedCount: 3,  
  upsertedCount: 0  
}
```

18: